

HW 6

1.

relation on $A = \{0, 1, 2, 3\}$ is

$$R = \{(0, 0), (0, 1), (3, 3), (3, 2)\}.$$

- (a) A relation on A is reflexive only if $(a, a) \in R$ for every $a \in A$, but $(1, 1)$ and $(2, 2)$ are not in R
- (b) A relation is irreflexive only if no element is related to itself but $(0, 0)$ and $(3, 3)$ are in R
- (c) A relation is symmetric only if $(x, y) \in R$ always implies $(y, x) \in R$, but $(0, 1) \in R$ while $(1, 0) \notin R$.
- (d) There are no distinct elements $x \neq y$ such that both $(x, y) \in R$ and $(y, x) \in R$.
- (e) Every possible chain in R closes correctly: $(0, 0)$ with $(0, 0)$ gives $(0, 0)$, $(0, 0)$ with $(0, 1)$ gives $(0, 1)$, $(3, 3)$ with $(3, 3)$ gives $(3, 3)$, and $(3, 3)$ with $(3, 2)$ gives $(3, 2)$; there are no other chains to check because 1 and 2 have no outgoing arrows.
- (f) An equivalence relation must be reflexive, symmetric, and transitive, but R is not reflexive and not symmetric.

2.

$A = \{3, 4, 5, 6, 7, 8\}$ and define R on A by

$$xRy \iff 3 \mid (x - y).$$

we prove that R is an equivalence relation

Reflexive

we have $x \in A$ then

$$x - x = 0.$$

since $3 \mid 0$ so we have xRx so R is reflexive

Symmetric

we assume xRy then $3 \mid (x - y)$ so there exists $k \in \mathbb{Z}$ where

$$x - y = 3k.$$

then

$$y - x = -(x - y) = -3k = 3(-k).$$

so $3 \mid (y - x)$, so yRx so R is symmetric

Transitive

lets assume xRy and yRz so $3 \mid (x - y)$ and $3 \mid (y - z)$, so there exist integers $k, \ell \in \mathbb{Z}$ where

$$x - y = 3k \quad \text{and} \quad y - z = 3\ell.$$

adding these equations gives

$$x - z = (x - y) + (y - z) = 3k + 3\ell = 3(k + \ell).$$

so $3 \mid (x - z)$, so xRz so therefore R is transitive

since R is reflexive, symmetric, and transitive, R is an equivalence relation

3.

(a)

$$\begin{aligned} [0] &= \{\dots, -6, -3, 0, 3, 6, 9, \dots\}, \\ [1] &= \{\dots, -5, -2, 1, 4, 7, 10, \dots\}, \\ [2] &= \{\dots, -4, -1, 2, 5, 8, 11, \dots\}, \\ [3] &= [0] = \{\dots, -6, -3, 0, 3, 6, 9, \dots\}, \\ [4] &= [1] = \{\dots, -5, -2, 1, 4, 7, 10, \dots\}, \\ [5] &= [2] = \{\dots, -4, -1, 2, 5, 8, 11, \dots\}. \end{aligned}$$

(b) every integer has remainder 0 or 1 or 2 when divided by 3, so every integer belongs to one of the classes $[0]$, $[1]$, or $[2]$, and these are all different

(c) two equivalence classes are either equal or disjoint and here the three distinct classes are $[0]$, $[1]$, and $[2]$ so

$$[0] \cap [1] = [0] \cap [2] = [1] \cap [2] = \emptyset.$$

their union is all integers:

$$[0] \cup [1] \cup [2] = \mathbb{Z}.$$

4.

so we want to prove:

$$R \subseteq S \text{ and } R \subseteq T \implies R \subseteq S \cap T.$$

since $R \subseteq S$ and $R \subseteq T$ so to show $R \subseteq S \cap T$ we let $(a, b) \in R$

since $R \subseteq S$ so $(a, b) \in S$ and since $R \subseteq T$ it also follows that $(a, b) \in T$

so (a, b) is in both S and T which means

$$(a, b) \in S \cap T$$

because (a, b) was an arbitrary element of R every element of R is in $S \cap T$ so

$$R \subseteq S \cap T$$

this is the proof